

10th International Conference on Air Transport – INAIR 2021, TOWARDS AVIATION REVIVAL

Military autonomous drones (UAVs) - from fantasy to reality. Legal and Ethical implications.

Anna Konert^{a*}, Tomasz Balcerzak^b

^a*Faculty of Law and Administration and Director of the Institute of Air and Space Law at Lazarski University in Warsaw, Poland*

^b*Adjunkt at the Institute of Air and Space Law at Lazarski University in Warsaw, Poland*

Abstract

Unmanned aerial vehicles, (UAVs), called drones, have recently greatly increased their role from simple surveillance and reconnaissance to increasingly controversial targeted killings. Autonomous drones raise important judicial and ethical issues about responsibility for unintentional harm which will be discussed in this paper. Autonomy refers to respect for human autonomy (in contrast with the autonomy of a drone) and includes the free choice of individuals and groups. In the context of drones used, the ethical principle of autonomy is translated into the human values, responsibility and trust. Some features of these unmanned combat aircraft (UAVs), such as flying at high altitudes, the ability to collect effective information and their impact on civilians, influence decisions on the legality of these weapons, and the distance between the operator of the aircraft and military targets also affects the principles of international law and humanitarian law. The paper will analyze the legal and ethical issues on using drones as a weapon.

© 2022 The Authors. Published by ELSEVIER B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 10th International Conference on Air Transport – INAIR 2021, TOWARDS AVIATION REVIVAL

Keywords: drone; drone operators; liability of a State; human rights; military drones; right to life

* Corresponding author.

E-mail address: a.konert@lazarski.edu.p

1. Introduction

Military autonomous drones (UAVs) can fly to a specific location, pick their own targets and kill without the assistance of a remote human operator. Therefore, the idea of a "killer robot" has moved from fantasy to reality.

Most people would probably be willing to understand "autonomous drones" as "smart technology", for example drones that can operate on the basis of a self-selected option (which in military terminology is referred to as "system initiative" or "full autonomy"). Such drones are programmed to equip them with a large number of alternative responses to the various challenges they may encounter in carrying out their missions. This is not science fiction - the technology is largely in place, although - to our knowledge - no approved autonomous drone systems are yet operational. The limiting factor is not the technology, but rather the political will to create and admit to possessing such a politically "sensitive" technology that would allow lethal machines to operate without direct human supervision.

Almost all academic studies in artificial intelligence acknowledge that no commonly accepted definition of Artificial Intelligence-AI exists, in part because of the diverse approaches to research in the field.

According to Weisgerber (Weisgerber, 2017) it's any artificial system that performs tasks under varying and unpredictable circumstances without significant human oversight, or that can learn from experience and improve performance when exposed to data sets. An artificial system developed in computer software, physical hardware, or other context that solves tasks requiring human-like perception, cognition, planning, learning, communication, or physical action. An artificial system designed to think or act like a human, including cognitive architectures and neural networks. An artificial system designed to act rationally, including an intelligent software agent or embodied robot that achieves goals using perception, planning, reasoning, learning, communicating, decision-making, and acting.

According to Ilachinski (Ilachinski, 2017) automated systems it's "a physical system that functions with no (or limited) human operator involvement, typically in structured and unchanging environments, and the system's performance is limited to the specific set of actions that it has been designed to accomplish... typically these are well-defined tasks that have predetermined responses according to simple scripted or rule-based prescriptions".

Autonomous Weapon System (Lethal Autonomous Weapon System-LAWS) according to US Department of Defense is "a weapon system that, once activated, can select and engage targets without further intervention by a human operator." US Department of Defense also has definition for robot as "a powered machine capable of executing a set of actions by direct human control, computer control, or a combination of both. At a minimum it is comprised of a platform, software, and a power source."

Understanding the relationships between these terms can be challenging, as they may be used interchangeably in the literature and definitions often conflict with one another. For example, some studies delineate between automated systems and autonomous systems based on the system's complexity, arguing that automated systems are strictly rule-based, while autonomous systems exhibit artificial intelligence. Some, including the Department of Defense, categorize autonomous weapon systems based not on the system's complexity, but rather on the type of function being executed without human intervention (e.g., target selection and engagement) (Scharre and Horowitz, 2015).

Still others describe AI as a means of automating cognitive tasks, with robotics automating physical tasks. This framework, however, may not be sufficient to describe how AI systems function, as such systems do not merely replicate human cognitive functions and often produce unanticipated outputs. In addition, a robot may be automated or autonomous and may or may not contain an AI algorithm.

Lethal Autonomous Weapon Systems (LAWS) are a special class of weapon systems that use sensor suites and computer algorithms to independently identify a target and employ an onboard weapon system to engage and destroy the target without manual human control of the system. Although these systems are not yet in widespread development, they could enable military operations in communications-degraded or -denied environments in which traditional systems may not be able to operate (Ilachinski, 2017).

The most dangerous way to use drones is their deliberate lethal use as a firearm. Remote pilotless drones-UAVs have become part of modern warfare. Some features of these unmanned combat aircraft (UAVs), such as flying at high altitudes, the ability to collect effective information and their impact on civilians, influence decisions on the

legality of these weapons, and the distance between the operator of the aircraft and military targets also affects the principles of international law and humanitarian law (Kuo and Warford, 2019).

Armed UAVs, controlled from a base often thousands of miles away, are sent to eliminate the target(s). The key advantage of using drones is that it allows the military to attack the enemy while minimizing the damage and casualties. However, the drone attacks often cause significant collateral damage, killing many innocent civilians with the intended target. In an effort to reduce the cost of operating UAVs, manufacturers have made drones more and more autonomous, reducing the need for human interaction and instruction to the drone. However, the increased autonomy of the military's weapons has raised serious ethical questions about the responsibility for the collateral damage of indiscriminate drone strikes. Removing humans from the decision making of UAVs during drone strikes raises ambiguity about who should be responsible for the consequences of drone strikes. Is it the robot, the programmer, or the military (Kuo and Warford, 2019)?

The world's first recorded case of an autonomous drone attacking humans took place in March 2020. The United Nations (UN) security report on Second Libyan Civil War. States that Libyan forces used the Turkish-made drones to "hunt down" and jam retreating enemy forces, preventing them from using their own drones (Mizokami, 2021).

The Article will analyze the legal and ethical issues on using drones as a "killer robot".

2. Autonomous military operations

Autonomous drones have no legal definition. There are advanced drones programmed with algorithms for countless human-defined courses of action to meet emerging challenges (Dyndal et al., 2017).

Drones were first used by military forces. Therefore, the drone warfare concept is not new. For years, they were used to carry out reconnaissance, target infrastructure and attack people. The U.S. in particular has used drones extensively to kill militants and destroy physical targets (Hernandez, 2021).

The US Department of Defense has used drones in nearly every military operation since the 1950s to provide reconnaissance, surveillance, and intelligence for enemy forces. Currently, it is estimated that nearly 100 countries use military drones (Karyoti, 2021). Equipped with the latest generation cameras, they provide an accurate topography of the terrain and are used in combat and rescue missions. Those with artificial intelligence communicate with soldiers and provide them with information about enemy movements on an ongoing basis.

Drones can transport heavier and heavier loads. Equipped with sets of anti-tank guided missiles, they target targets and also help in developing war tactics.

The possible consequences of the use of unmanned aerial systems (UAV) have been reported, among others, by the press in connection with the inadvertent killing of two innocent hostages of American and Italian nationality during an attack carried out with a drone aimed at al-Qaeda warriors in 2015.

During the "Spring Shield" (March 2020) operation in Idlib against the Syrian troops, the Turkish armed forces made use of unmanned systems on an unprecedented scale. The appropriate doctrine of their use, also for combat tasks, and cooperation with other means of combat (including artillery) contributed to the high effectiveness of the activities of the Turkish armed forces.

Azerbaijan used armed drones to gain a major advantage over Armenia in recent fighting for control of the Nagorno-Karabakh region. In 2021 the Israel Defense Forces reportedly used drones to drop tear gas on protesters in the occupied West Bank, while Hamas launched loitering munitions — so-called kamikaze drones — into Israel (Hernandez, 2021).

3. Discussion on Legal and Ethical implications of using autonomous military drones

Technologies create some moral accountability gaps and such potential responsibility gaps must be addressed properly through technical solutions and legal regulations (Dyndal et al., 2017).

There are several views on the responsibility of autonomous weapon systems. In "The Case for Ethical Autonomy in Unmanned Systems" by roboticist Ronald C. Arkin (Arkin, 2020), Arkin argues a computer would struggle to operate. Modern artificial intelligence is still unable to accurately model the psychological tools of human thinking that allow humans to be creative and recognize patterns. For example, fully autonomous weapons systems are unable to distinguish combatants from non-combatants, while this is not too difficult for humans to do.

One of the biggest challenges to the development and approval of aircraft equipped with such technology is that it is extremely difficult to build a satisfactory authorization system that ensures that the technology is safe and works the way people would like it to. In practice, such 'smart' drones would need to be programmed with an incredible number of combinations of alternative workflows, which would make them impossible to verify and test at the level we apply to manned aircraft.

The other, more sensible view, called the Human-Machine Partnership approach, advocates an intermediate level of autonomy, where humans and artificial intelligence work together, combining human creativity with the unclouded and unemotional decisions of the computer. Limitations in current computer technologies make it difficult for fully autonomous UAVs to accurately distinguish between hostile and friendly. However, humans are equally poor at making sound decisions during war, evidenced by the numerous war crimes that occur during every war. Thus, with human and machine sharing control of the weapon system, the human then has the responsibility to act within the ethical boundaries of war as laid out in international war laws and treaties. In the human-machine partnership, there is no moral ambiguity, as the actions of the drone falls on the human operator.

The Department of US Defense's Unmanned Systems Roadmap: 2007-2032 provides additional reasons for pursuing autonomous weapons systems. These include that robots are better suited than humans for "dull, dirty, or dangerous missions" (Etzioni and Etzioni, 2017). An example of a dull mission is long-duration sorties. An example of a dirty mission is one that exposes humans to potentially harmful radiological material. An example of a dangerous mission is explosive ordnance disposal (Ryan, 2018).

In addition, the long-term savings that could be achieved through fielding an army of military robots have been highlighted. US Department of Defense figures showing that "each soldier in Afghanistan costs the Pentagon roughly \$850,000 per year (Francis, 2017). Conversely, according to Francis, "the TALON robot—a small rover that can be outfitted with weapons, costs \$230,000.

The other thing is that the physical strain of high-G maneuvers and the intense mental concentration and situational awareness required of fighter pilots make them very prone to fatigue and exhaustion; robot pilots, on the other hand would not be subject to these physiological and mental constraints. Moreover, fully autonomous planes could be programmed to take genuinely random and unpredictable action that could confuse an opponent.

The Case for Full Autonomy in Drones Proponents of fully autonomous weapon systems, notably Ronald Arkin (Arkin, 2020), argue that current and future developments in unmanned weapons systems allow them to behave even more ethically than humans when fighting. If engineers can create drones that can ethically outperform humans, then it should be pursued, even if the fully autonomous system is still imperfect. A fully autonomous weapon system can greatly reduce collateral damage and property damage. Arkin contends that robots are already faster, smarter, and stronger than humans, so it is not outrageous to suggest that robots can act more ethically than humans. It is already difficult enough for humans to distinguish between combatants and non-combatants, but future fully autonomous systems may be able to outperform humans in all aspects of combat, for these five following reasons.

First, robots do not need to protect themselves when the target is unclear or indistinguishable from innocent civilians. For autonomous robots, self-preservation is not a high priority, or a priority at all; military officers can use fully autonomous robots in a self-sacrificing that UAVs should be given full autonomy to prevent war crimes. Autonomous weapon systems are free from human emotions that prompt them to commit war crimes. However, while a valid point, current artificial intelligence technology is unable to replicate specific human experiences and capabilities. Completely removing humans from UAVs would severely underutilize humans ability to use past experiences and fundamental knowledge to deal with unfamiliar situations that a computer would struggle to operate. Modern artificial intelligence is still unable to accurately model the psychological tools of human thinking that allow humans to be creative and recognize patterns. For example, fully autonomous weapons systems are unable to distinguish combatants from non-combatants, while this is not too difficult for humans to do.

The other, more sensible view, called the Human-Machine Partnership approach (Arkin, 2020) advocates an intermediate level of autonomy, where humans and artificial intelligence work together, combining human creativity with the unclouded and unemotional decisions of the computer. Limitations in current computer technologies make it difficult for fully autonomous UAVs to accurately distinguish between hostile and friendly. However, humans are equally poor at making sound decisions during war, evidenced by the numerous war crimes that occur during every war. Thus, with human and machine sharing control of the weapon system, the human then has the responsibility to

act within the ethical boundaries of war as laid out in international war laws and treaties. In the human-machine partnership, there is no moral ambiguity, as the actions of the drone falls on the human operator.

Human-robot interaction refers to “how people work or play with robots” (Scharre and Horowitz, 2015). Robots are quite different from other computers or tools because they are “physically situated agents,” and human users interact with them in distinct ways (Scharre and Horowitz, 2015). Research on interaction needs to span a number of domains well beyond engineering, including psychology, cognitive science, and communications, among others.

It would be dangerous to have weapon systems be fully autonomous, at least with the current level of artificial intelligence technologies. Computer programs still have challenge in accurately model unique human behaviors such as creativity and pattern recognition. The human ability to react and perform under unexpected circumstances gives humans a distinct advantage over computer-controlled weapons, especially right now. Because of the current limitations of computer technology, it would be foolhardy for the military to implement fully autonomous weapon systems.

So, it might be solution to have human-machine partnership. The human-machine partnership approach is a much more sensible alternative to fully autonomous weapons and is much easier to implement. The human-machine partnership combines the creativity of human operators with the advanced data processing and data sensors of the computers in the drones. This partnership divides the labor and responsibility of operating the drones between the human and the computer. The computer is in control of the data collection and processing, while the human is in charge of more important decisions, like identifying targets and distinguishing items where there is low visibility. When a human is involved with the operation of drones, the operator is always aware of the situation surrounding the drone. This also gives a clear responsibility for the actions of the drone, with fully autonomous systems the responsibility is very unclear. In the human-machine partnership approach, the human and machine complement each other. Humans and computers can develop different perspectives of a particular situation. The human operator can use the extra information gathered and processed by the computer to better inform a decision about the situation. For example, a human operator on a naval vessel could use the computer’s advanced sensors to identify the objects surrounding the vessel, and the operator could identify the ambiguous objects. The partnership approach gives the drone more autonomy, removing the human operator and reducing operating costs, but leaves critical decisions, such as whether or not to terminate a target, to a human.

For the human-partnership approach, the human is clearly responsible for the actions of the drone. This creates an ethical boundary agent for the operator, forcing him/her to act within the ethical boundaries of combat the ultimate decision on whether or not to carry out the strike. For example, a drone is flying a mission and identifies the hostiles on the ground and alerts the human operator that the targets have been spotted. This takes advantage of the superior sensors and data processing on the drone, which are used to identify targets where humans cannot see. However, the human operator may also identify structures, such as hospitals or civilian buildings nearby, where the computer cannot differentiate between hostile and friendly buildings. It is the responsibility of the human operator to use all the advanced information provided by the drone to decide whether it is ethically permissible to carry out the strike. In this scenario, there is no ethical ambiguity for who is responsible for actions of the drone. The drone gathers and processes information and carries out orders all within the mission parameters and all without human supervision. However, essential mission decisions, especially the use of lethal force, are left for the human to decide. The drone operator assumes full responsibility of the actions and consequences of the drone.

While the human-machine partnership is a more sensible approach to maximize the strengths of both humans and robots in drones, the partnership approach still has significant shortcomings. Proponents of fully autonomous weapon systems are correct in claiming that humans are embarrassingly bad at making ethically responsible decisions in the fog of war (Kuo and Warford, 2019). In addition, the use of drones creates an emotional gap between the human and the victim, and can cause even further dehumanization of the victims, creating greater risk for war crimes to be committed by the human operator. In the human-machine partnership approach, it is the duty of the human operator to make ethically responsible decisions about the mission, given the enhanced intelligence provided by the advanced sensors and data collected by the drone. As technology develops, it is ideal for drones to become fully autonomous. Once it is possible to model human behavior and decision-making, drones will be able to make ethical decisions. Robots, with the right technology, will be able to perform more ethically than humans, drastically reducing the war crimes and unethical actions during war (Kuo and Warford, 2019).

It is difficult to argue that autonomous drones could theoretically meet the “*ius in bello*” (law in war) criterion that would allow for the distinction of military actions in line with the tradition of “just war”. Moral judgments about who may legitimately be targeted under combat conditions are difficult even for normal soldiers. The concern is that allowing autonomous drones to make such distinctions would most likely result in civilian casualties and unacceptable collateral damage. Even if such a weapon system could distinguish between soldiers and non-combatants, it is still questionable whether an autonomous drone could judge the proportionality of an attack - that is, whether an attack would cause unnecessary suffering. However, going beyond the uncertainty as to whether autonomous drones will gain the ability to make such distinctions in the future, it could also be argued that if such weapons systems were not able to operate as required by “*ius in bello*”, they are unlikely to be deployed especially in operational conditions where there is a high risk of causing undue harm to the civilian population.

4. The right to life on the basis of the European Convention for the Protection of Human Rights and Fundamental Freedoms.

A global survey commissioned by the Campaign to Stop Killer Robots in 2020 found that a majority of respondents — 62% — said they opposed the use of lethal autonomous weapons systems (Hernandez, 2021). There is a general fear of civilians and especially in the territories with armed conflicts.

Autonomous drones used during armed conflict, would be subject to the general principles and rules of the Law of Armed Conflict and therefore they are not to be distinguished from any other weapons, weapon systems or weapon platforms. As with any “means of warfare”, autonomous drones must only be directed at lawful targets (military objectives and combatants) and attacks must not be expected to cause excessive collateral damage (Dyndal et al., 2017).

Some features of these unmanned combat aircraft (UAVs), such as flying at high altitudes, the ability to collect effective information and their impact on civilians, influence decisions on the legality of these weapons, and the distance between the operator of the aircraft and military targets also affects the principles of international law and humanitarian law. The use of drones has sparked a major debate about the legality of such targeted killings under international law. Assuming that the use of drones to attack suspected terrorists and in armed conflict violates international law, do actual or potential victims have an opportunity to obtain a compensation? In addition, drones are also used outside the context of an armed conflict. In order to assess the legality and legitimacy of deliberate killings by armed drones, a distinction should be made between an armed conflict situation and a non-armed conflict situation (law enforcement). The rules of redress outside the context of an armed conflict are largely dictated by international public law, especially human rights.

Article 2 of the European Convention on Human Rights ranks as one of the most fundamental provisions in the Convention, one which in peace time, admits of no derogation under Article 15. It states that “*1. Everyone’s right to life shall be protected by law. No one shall be deprived of his life intentionally save in the execution of a sentence of a court following his conviction of a crime for which this penalty is provided by law. 2. Deprivation of life shall not be regarded as inflicted in contravention of this article when it results from the use of force which is no more than absolutely necessary: (a) in defense of any person from unlawful violence; b) in order to effect a lawful arrest or to prevent the escape of a person lawfully detained; (c) in action lawfully taken for the purpose of quelling a riot or insurrection.*”

As stated in the Guide on Article 2 of the European Convention on Human Rights “Right to life”, the ECHR approach to the interpretation of Article 2 must be guided by the fact that the object and purpose of the Convention as an instrument for the protection of individual human beings requires that its provisions must be interpreted and applied so as to make its safeguards practical and effective (*McCann and Others v. the United Kingdom*, § 146).

The jurisprudence of the European Court of Human Rights (ECHR) announced the criteria for the use of force in the framework of the so-called law enforcement. A necessary condition for killing a person is that he poses a concrete threat to human life, and at the same time is suspected of having committed a serious crime. Thus, the deliberate killing of a person can only be justified by the protection of human life. The most decisive criterion for the use of drones in law enforcement, e.g. to eliminate terrorist suspects, is reflected in Principle of the UN's 9 Basic Principles on the Use of Force and Firearms by Law Enforcement Officers, according to which officers may not use firearms against individuals, with the exception of self-defense or the defense of other persons against an immediate

threat to life or serious bodily injury, in order to prevent the commission of a particularly serious crime posing a serious threat to life, arrest a person or prevent his escape, and only when less extreme measures are insufficient to achieving these goals. In any case, the deliberate, lethal use of a firearm can only be done when it is absolutely inevitable to save life (Konert, 2020).

5. Conclusions

The rapid development and increased use of drones have raised serious ethical questions about their use. As the military pushes for greater autonomy in drones in order to reduce operating costs, the responsibility for the actions of the drone becomes ambiguous. There are a wide range of views on how to ensure drones act ethically, ranging from mistrust to advocating full autonomy for drones. The unethical behavior of humans in war is largely a result of human emotions clouding their judgment, resulting in an unethical decision. Robots are unaffected by human emotions such as anger and desire for revenge, and they possess advanced sensors and data processing that allows them to be more effective and ethical than humans in combat. However, current robot technologies are limited, and the technology does not yet exist for drones to evaluate situations and make an ethical decision faster or better than humans.

Human behavior and decision-making is very difficult to model and program into a drone. Thus, currently, it is more sensible to establish a human-machine partnership, where humans work with robots and maximize effectiveness by combining the strengths of both partners. The drone will not require a human operator, as they do now, and will carry out their orders and make non-critical decisions. However, critical decisions, especially the use of lethal force, are made by a human. The human can use the advanced sensors on the drone to gather information about the mission situation and make the final whether or not to engage the targets. Until artificial intelligence technology progresses to make fully autonomous drones feasible, humans and robots will have to work together. However, in order to further reduce the loss of human lives in combat, drones should become fully autonomous. With the ability to distinguish civilian from hostile, drones cannot commit war crimes out of anger, but will perform more ethically than humans.

Although AI has not yet entered the combat arena in a serious way, experts are predicting the potential impact that AI will have on the future of warfare. This influence will be a function of many factors, including the rate of commercial investment, the drive to compete with international rivals, the research community's ability to advance the state of AI capability, the military's general attitude toward AI applications, and the development of AI-specific warfighting concepts.

While many analysts admit that military AI technology is in a stage of infancy, it is difficult to find an expert who believes that AI will be inconsequential in the long run. However, AI critics point to a number of trends that may minimize the technology's impact. From a technical standpoint, there is a potential that the current safety problems with AI will be insurmountable and will make AI unsuitable for military applications.

The vision of a world where decisions about life and death will one day be made by machines not under human direct control must be considered seriously. The last few years have seen a rapid development in the field of drone technology, the level of autonomy of which is constantly increasing. Although, as far as we know, no approved autonomous drone systems have achieved operational capability, the technology is currently being tested and developed. Some see new opportunities and potential benefits of using autonomous drones. Others view the development and use of such technology as immoral in principle. Influential people such as Stephen Hawking, Elon Musk and Steve Wozniak have already urgently called for a ban on hostilities using autonomous weapons or artificial intelligence.

There is no doubt that autonomous drones pose important problems for the judiciary and ethics related to liability for damage. These technologies create some gaps in the moral and legal responsibility system. Such potential liability gaps must be properly addressed through technical solutions and legal regulations.

Acknowledgement

The Article was prepared under the National Science Centre fellowship No UMO-2017/27/B/HS5/00008.

References

- Arkin, R. C., 2020. Altmetric Articles The Case for Ethical Autonomy in Unmanned Systems, The Case for Ethical Autonomy in Unmanned Systems, <https://www.tandfonline.com/doi/abs/10.1080/15027570.2010.536402>
- Dyndal, L., Berntsen, A., Redse-Johansen, S., 2017. Autonomous military drones: no longer science fiction, <https://www.nato.int/docu/review/articles/2017/07/28/autonomous-military-drones-no-longer-science-fiction/index.html>
- Etzioni, A., Etzioni, O., 2017. Pros and Cons of Autonomous Weapons Systems, <https://www.armyupress.army.mil/Journals/Military-Review/English-Edition-Archives/May-June-2017/Pros-and-Cons-of-Autonomous-Weapons-Systems/>
- Francis D., 2017. How a New Army of Robots Can Cut the Defense Budget, <http://www.thefiscaltimes.com/Articles/2013/04/02/Howa-New-Army-of-Robots-Can-Cut-the-Defense-Budget>.
- Hernandez, J., 2021. A Military Drone With A Mind Of Its Own Was Used In Combat, U.N. Says, <https://www.npr.org/2021/06/01/1002196245/a-u-n-report-suggests-libya-saw-the-first-battlefield-killing-by-an-autonomous-d?t=1625127946959>
- Ilachinski, A., 2017. AI, Robots, and Swarms: Issues, Questions, and Recommended Studies, Center for Naval Analysis, January 2017, p. 6.
- Karyoti, V., 2021. 9/11's legacy of drone warfare has changed how we view the military, <https://theconversation.com/9-11s-legacy-of-drone-warfare-has-changed-how-we-view-the-military-167393>
- Konert, A., 2020. Unmanned Aircraft Systems. A New Era of Air Law. Private law issues, C.H.Beck, Warsaw.
- Kuo I., Warford, E., 2019, The Ethical Performance of Drones, University of Washington School of Law
- Mizokami, K., 2021. For the First Time, Drones Autonomously Attacked Humans. This Is a Turning Point, <https://www.popularmechanics.com/military/weapons/a36559508/drones-autonomously-attacked-humans-libya-united-nations-report/>
- Ryan, M., 2018. Integrating Humans and Machines, The Strategy Bridge, January 2, 2018, <https://thestrategybridge.org/the-bridge/2018/1/2/integrating-humans-and-machines>
- Scharre, P., Horowitz, M., 2015. An Introduction to Autonomy in Weapon Systems, Center for a New American Security, February 2015, pp. 6-7.
- Weisgerber, M., 2017. The Pentagon's New Algorithmic Warfare Cell Gets Its First Mission: Hunt ISIS, Defense One, May 14, 2017, <http://www.defenseone.com/technology/2017/05/pentagons-new-algorithmic-warfare-cell-gets-itsfirst-mission-hunt-isis/137833/>